

3.4 Desertification

1. Environments Vulnerable to Desertification

Desertification does not happen in the desert — there is nothing left there to degrade. It happens in the **semi-arid and dry sub-humid land at the desert margin**: the Sahel south of the Sahara, the drylands of southern Africa, north-east Brazil, western India and Pakistan, central Asia, inland Australia and the American south-west. These regions are vulnerable precisely because they receive enough rain to support farming and grazing in a good year, but not reliably enough to sustain that use every year.

Understanding where they are means first understanding where the true deserts are, because the vulnerable belt wraps around them. Hot deserts are found predominantly at approximately **30° North and 30° South** of the equator, in belts corresponding to the **subtropical high pressure zones**. They also occur in **rain shadows** (sheltered side of mountain ranges) and deep **continental interiors** far from moisture sources.

Desert Definition: An area receiving less than **250 mm of rainfall per year**. Hot deserts experience extreme heat by day and cold at night due to the absence of cloud cover and low humidity.

Climate Characteristics

Variable	Value	Detail
Annual rainfall	<250 mm/year (often <25 mm)	Rain is rare, unpredictable, and often arrives as intense short bursts causing flash floods
Daytime temperature	40–50°C in summer	No cloud cover means intense solar radiation reaches the ground directly

Night-time temperature	Can drop to 5–10°C	No cloud cover also means heat is rapidly lost at night via radiation — no "blanket" effect
Diurnal temperature range	Up to 40°C difference	Very large daily swing — extreme physical weathering (exfoliation/insolation weathering)
Humidity	Very low (<20%)	Low humidity accelerates evaporation; any moisture is rapidly lost to the atmosphere

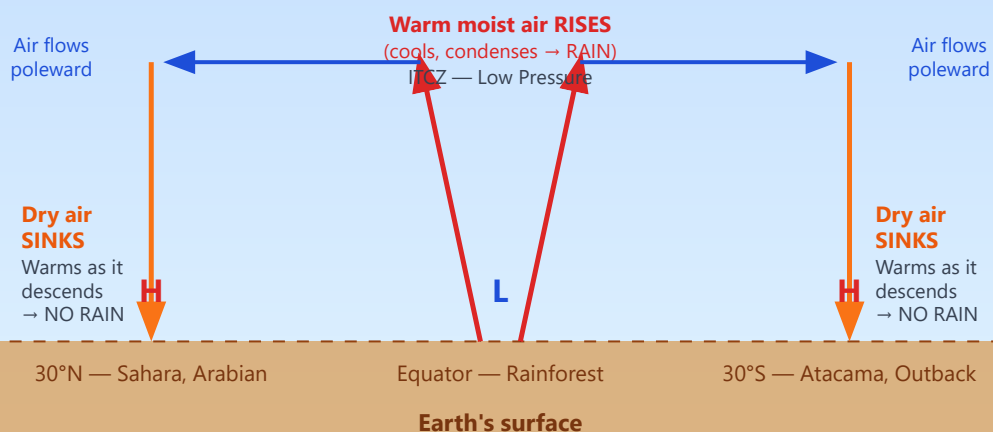
Major Hot Desert Examples

Desert	Location	Notable Feature
Sahara	North Africa	World's largest hot desert (~9.2 million km ²); only ~25% is sand — rest is rock (reg/hammada)
Arabian	Arabian Peninsula (Saudi Arabia, Oman, UAE)	World's largest sand desert (Rub' al Khali — Empty Quarter)
Atacama	West coast South America (Chile, Peru)	World's driest — some areas have had no recorded rainfall for centuries
Sonoran	SW USA and NW Mexico	Home of the saguaro cactus; most biodiverse desert
Australian Outback	Central Australia	Covers ~70% of Australia's land area

2. Why These Environments Are Dry — Global Circulation and High Pressure

The location of hot deserts at ~30°N/S is explained by the **Hadley Cell** — a large atmospheric circulation pattern driven by the sun heating the equator.

The Hadley Cell — Why Deserts form at 30°N/S



Other Causes of Desert Climate

Cause	Mechanism	Example
Subtropical high pressure	Hadley cell — dry air descends at 30°N/S, warms adiabatically, inhibits cloud formation and rainfall	Sahara, Arabian, Thar deserts
Rain shadow effect	Moist air rises over mountains, deposits rain on windward side; descends dry and warm on leeward side	Atacama Desert (behind the Andes); Patagonian Desert (behind Andes)
Distance from sea (continentality)	Air masses lose moisture as they travel inland; no maritime moisture source	Gobi Desert (Central Asia)
Cold ocean currents	Cold water chills overlying air, preventing it from rising and forming rain clouds; stable cold air = no convection	Atacama — cooled by the Humboldt Current; Namib — Benguela Current

3. A Changing Distribution — Climatic Patterns Over Decades

The specification asks not just where vulnerable environments are, but **how that pattern is changing over time**. Desertification is not the Sahara advancing as a solid front; it is patches of degraded land appearing, spreading and sometimes recovering within the semi-arid belt.

Unreliable Rainfall

The defining climatic feature of a semi-arid region is not that rainfall is low but that it is **variable**. A region with a long-term average of 400 mm may receive 700 mm one year and 150 mm the next. Two consequences follow:

- **Wet runs encourage expansion.** During a decade of above-average rainfall, farmers and herders move onto land at the desert margin that seems productive. When rainfall returns to normal — or falls below it — that land cannot support the population now depending on it, and it degrades quickly.
- **Averages hide the risk.** A long-term mean tells you very little about whether a crop will succeed in any given year. Planning based on averages rather than on variability is one of the commonest forms of poor land management.

The Sahel provides the clearest example: a long run of relatively wet years in the 1950s and early 1960s was followed by a severe multi-decade drying from the late 1960s through the 1980s, and then by a partial recovery of rainfall since the 1990s. The vegetation and the livelihoods did not simply recover with the rain, because by then the soil itself had been lost in many places.

Higher Rates of Evapotranspiration

Rising temperatures increase **potential evapotranspiration** — the amount of water the atmosphere can take up from soil and plants. Even where rainfall totals hold steady, higher temperatures mean:

- More of each rainfall event is evaporated before it can infiltrate and reach roots.
- Soils dry out faster between showers, shortening the effective growing season.
- The **aridity index** — the ratio of rainfall to potential evapotranspiration — falls, so a region can become functionally drier without receiving less rain.

This is why the belt of land classed as vulnerable to desertification has been expanding at its poleward and equatorward edges, and why the UNCCD treats climate change and land degradation as a single problem rather than two.

Exam Tip — "Changing Over Time": If a question asks how the distribution of vulnerable environments is changing, do not simply say "it is getting bigger". Explain the mechanism: rainfall is becoming less reliable and evapotranspiration is rising, so the boundary between semi-arid and arid conditions migrates, and land that was marginal becomes unusable. Name the Sahel and give the decadal pattern.

4. Smaller-Scale Processes — Vegetation, Evapotranspiration and Micro-climate

Desertification is driven by global circulation at the large scale, but the actual damage happens at the scale of a single field or a few hundred metres of grazing land. Three linked small-scale processes do the work.

Changing Patterns of Vegetation

Vegetation is the protective layer between the atmosphere and the soil. As cover thins:

- **Interception falls.** Rain reaches the ground at full force. Raindrop impact breaks up soil crumbs and the fine particles seal the surface pores — a process called **surface capping**. Infiltration can drop by more than half.
- **Runoff rises.** Water that cannot infiltrate flows overland, carrying topsoil into gullies. The water is lost to the local system entirely rather than being stored in the soil.
- **Root binding is lost.** Without roots to hold aggregates together, dry soil is exposed to wind deflation during the dry season.
- **Species composition shifts.** Palatable perennial grasses are grazed out first and are replaced by annuals, unpalatable species and woody scrub, which give far less continuous ground cover.

Evapotranspiration and the Moisture Feedback

Plants return large volumes of water to the atmosphere through transpiration. Remove them and:

1. Evapotranspiration falls, so less moisture is recycled into the air above the region.
2. Less atmospheric moisture means fewer clouds and less convectional rainfall downwind.
3. Less rain means still less vegetation — a **positive feedback loop** that makes degradation self-sustaining.

Bare ground also has a higher **albedo** than vegetated ground: it reflects more solar radiation, so less energy is available to warm the near-surface air and drive the uplift that produces convectional rain. This is the Charney mechanism, and it is one reason degraded land tends to stay degraded even when regional rainfall recovers.

Micro-climate

At the scale of a few metres, vegetation creates its own climate, and losing it changes conditions sharply:

Factor	Under vegetation cover	On bare ground
Surface temperature	Shaded; soil stays cooler by day	Full insolation; surface temperatures can exceed 60°C, killing seedlings and soil organisms
Wind speed at the surface	Reduced by plant stems and tree canopies	Unimpeded; fine particles are lifted and removed
Soil moisture	Retained by shade, litter and organic matter	Evaporated rapidly; soil dries within hours of rain
Organic matter	Continually replenished by leaf and root litter	Oxidised and lost; soil loses its ability to hold water

This is why almost every successful anti-desertification technique is really a micro-climate intervention: a stone line, a pit, a windbreak or a retained tree changes conditions over a few square metres, and a landscape is restored a few square metres at a time.

Vegetation Adapted to These Conditions

Xerophyte: A plant with structural or physiological adaptations that allow it to survive where water is very scarce — from the Greek *xero* (dry) and *phyte* (plant). Deep tap roots, waxy cuticles, reduced leaves, water storage tissue and seeds that germinate only above a rainfall threshold are all xerophytic adaptations.

Understanding these adaptations matters for management: **species that already survive locally are almost always a better basis for restoration than introduced species**, which is the single clearest lesson

from the early years of the Great Green Wall, where large numbers of planted seedlings died because they were placed where the rainfall could not sustain them.

5. Human Activities That Contribute to Desertification

Desertification: The process by which fertile land in **dryland areas (arid, semi-arid, and dry sub-humid)** gradually degrades and becomes desert-like. It involves loss of topsoil, vegetation, and land productivity. Defined by the **UNCCD** (United Nations Convention to Combat Desertification). Affects approximately 1.5 billion people globally and threatens ~40% of Earth's land surface.

The human activities below do not act alone. Each one removes vegetation or soil moisture, and it is the *combination* of human pressure with the unreliable rainfall described in section 3 that tips land over into degradation. The specification names four in particular: **unsustainable use of water resources, over-grazing, poor land management and the use of fire wood.**

Causes of Desertification

Cause	Mechanism	Human or Physical
Overgrazing	Too many livestock (cattle, goats, camels) consume vegetation faster than it can regenerate; roots die; bare soil exposed to wind erosion; soil compacted by hooves	Human (population growth, cattle as status symbol)
Deforestation (fuelwood collection)	Trees and shrubs cut for firewood (80% of sub-Saharan Africa relies on wood for cooking fuel); roots no longer hold soil; wind erosion increases	Human (poverty, lack of alternatives)
Over-cultivation	Land farmed without rotation or fallow periods; soil nutrients exhausted; soil structure degrades; wind and rain erode bare soil between crops	Human (food demand, poverty)
Population growth	More people = more demand for food and fuel from same fixed land area; fragile marginal land brought into cultivation; pressure increases	Human

Climate change / drought	Reduced and increasingly erratic rainfall across the Sahel (linked to warming Indian Ocean and shifts in ITCZ); longer drought periods mean vegetation cannot recover between dry seasons	Physical (exacerbated by human climate change)
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Effects of Desertification

- **Soil erosion:** Topsoil blown away by wind or washed by flash floods, exposing infertile subsoil
- **Reduced crop yields:** Degraded soil cannot support adequate harvests; food insecurity worsens
- **Famine and malnutrition:** Crop failures lead to starvation — particularly affecting children under 5
- **Migration:** People abandon degraded land and move to urban areas or other countries (climate/environmental refugees)
- **Conflict:** Competition for remaining productive land and water resources between farmers and herders has intensified conflicts (e.g. Sahel violence in Burkina Faso, Mali, Niger)
- **Economic loss:** UNCCD estimates desertification costs ~\$42 billion per year in lost economic output

6. Stakeholders — Viewpoints, Values and Attitudes

The specification asks explicitly about **the different viewpoints, values and attitudes of stakeholders in the region**. Desertification is rarely caused by anyone behaving unreasonably. It is usually the aggregate result of several groups each making a rational decision for themselves on land that none of them controls alone.

Stakeholder	Priority and viewpoint	Where this creates conflict
Pastoralists (e.g. Fulani, Tuareg)	Need mobility and access to seasonal grazing and water. Herd size is both wealth and insurance against drought, so reducing it voluntarily is a serious risk	Clashes with farmers whose fields block traditional migration routes, and with governments that want settled, taxable, countable populations
Settled arable farmers	Need secure land tenure and to extend cultivation to feed growing families. Fencing protects crops from livestock	Expansion onto grazing land removes pastoralists' dry-season reserve; fences sever migration corridors

Women	In most Sahelian households, women collect fuelwood and water. Longer collection journeys directly cost them time and education	Often excluded from land tenure decisions despite bearing the heaviest costs of degradation; interventions designed without them frequently fail
National governments	Want food security, export earnings, revenue and political stability. Often favour large, visible projects that attract international funding	Cash-crop schemes on marginal land and subsidised boreholes can accelerate degradation; top-down projects may ignore local practice
NGOs	Focus on local, low-cost, community-owned techniques with measurable outcomes	Work at a scale far smaller than the problem, and depend on donor funding cycles that are shorter than ecological recovery times
International bodies (UNCCD, African Union, donors)	Frame desertification as a global problem of climate, migration and security	Pledges are slow to convert into work on the ground; targets set internationally may not match local priorities
Agribusiness and mining companies	Seek land, water and minerals for profit, often on long leases	Large-scale irrigation and extraction can draw down groundwater that local users depend on

The tragedy of the commons: Where a resource such as grazing land is shared and unregulated, it is rational for each individual user to add one more animal, because they receive the whole benefit while the cost of the extra grazing is spread across everyone. If every user reasons the same way, the resource is destroyed. This explains why appeals to individual restraint fail and why successful management almost always involves an agreed, enforced collective rule.

Exam Tip — Writing About Conflicting Attitudes: Avoid casting any group as simply careless. The mark schemes reward answers that show *why* a damaging decision is rational for the person making it — a herder cannot reduce his herd because it is his only drought insurance; a farmer cannot leave land fallow because his family needs food this year. Then explain what would have to change for the decision to be different: secure tenure, an alternative income, an enforceable collective agreement.

7. Case Study: Desertification in the Sahel

Case Study: Desertification in the Sahel

Location: A belt of semi-arid land approximately 350–500 km wide stretching ~5,400 km across Africa: Mauritania, Senegal, Mali, Burkina Faso, Niger, Chad, Sudan, South Sudan, Ethiopia, Eritrea.

Population: Well over 100 million people depend on this region for subsistence agriculture and pastoralism. Population is growing at ~3% per year — among the world's fastest rates — increasing pressure on land dramatically.

Historical famines:

- 1968–1973: Prolonged drought; ~100,000 deaths; millions displaced
- 1984–1985: Worst drought of the 20th century in Ethiopia/Sudan; hundreds of thousands of deaths (estimates up to 1 million); Live Aid response
- 2011–2012: Horn of Africa drought declared a famine; 260,000 deaths in Somalia alone
- 2021–2023: Ongoing food crisis with tens of millions of people facing acute food insecurity in the Sahel

The Great Green Wall project: A pan-African initiative launched in 2007 by the African Union — a planned 8,000 km × 15 km belt of native trees and vegetation stretching from Senegal to Djibouti, designed to combat desertification, improve food security, and sequester carbon (250 million tonnes by 2030).

- Progress: roughly 18 million hectares restored and about 350,000 jobs created, with over US\$14 billion raised or pledged — the initiative's own assessment puts it at around 15% of its 2030 goal
- Challenges: funding that is pledged faster than it is spent, political instability in the Sahel (multiple coups in Burkina Faso, Mali and Niger between 2020 and 2023), drought, maintenance after planting, and conflict making access impossible in some areas
- Successes: in Senegal and Ethiopia, restored areas show improved soil health, food production and biodiversity
- Full evaluation in section 9 below

Water harvesting — Zai pits (Burkina Faso): Small pits (20–30 cm diameter, 20cm deep) dug in rows across degraded farmland. Fill with compost/manure. Rainwater collects in pits, allowing crops to be grown where soil is too hard for rain to penetrate. Revived and improved by farmer Yacouba

Sawadogo from the 1980s, whose methods spread widely across Burkina Faso and Niger. A proven low-tech, low-cost solution.

Drought-resistant crops: NGOs and the CGIAR research network have developed drought-tolerant varieties of sorghum, millet, and cowpea. WEMA (Water Efficient Maize for Africa) project distributes drought-tolerant maize seeds to small farmers.

8. Local Strategies — NGOs Working With Communities

Strategy Overview

Strategy	Detail	Evaluation
Afforestation	Planting trees to stabilise soil, reduce wind speed, and restore water cycle through evapotranspiration	Effective long-term; takes years to establish; needs maintenance and water; wrong species choices can fail
Stone lines / bunds ("magic stones")	Rows of stones laid along contour lines to slow water runoff and encourage infiltration; retains soil during rare rainstorms	Simple, cheap, can be built by local communities; well-evidenced in Burkina Faso to increase crop yields by 50%; see detail below
Zai pits	Traditional technique (see case study above) — pits with organic matter concentrate water and nutrients for crops in degraded soil	Proven highly effective; low-cost; community-led; Yacouba Sawadogo was awarded the "Right Livelihood Award" in 2018
Reduced stocking density	Limiting the number of livestock per hectare to sustainable levels; allows vegetation to recover between grazing periods	Difficult to enforce in nomadic cultures where livestock = wealth; requires community agreement and often land tenure reform

Drought-tolerant crops	Breeding or GM varieties that require less water and mature faster (before drought intensifies)	Effective yield improvement; some opposition to GM crops; needs seed distribution networks to reach smallholders
International aid and agreements	WFP provides emergency food; UNCCD coordinates global anti-desertification policy; Great Green Wall funding	Emergency aid addresses symptoms, not causes; long-term development investment more valuable; UNCCD has limited enforcement power
FMNR (see below)	Farmer Managed Natural Regeneration — encouraging existing tree stumps and roots to regenerate rather than planting new trees	Faster, cheaper, and more ecologically appropriate than afforestation; transformational results in Niger — see below

Stone Bunds ("Magic Stones") — Evidence and Detail

Stone bunds are simple but highly effective contour barriers. The technique involves:

1. Identifying the contour lines of a degraded slope (using a simple A-frame level made from two sticks and a string)
2. Collecting rocks from the surrounding area and arranging them in lines following the contour — typically stones 10–20cm in diameter, placed in rows 20–30cm high
3. Rainwater flowing down the slope is slowed and pooled behind the stone lines, allowing it to infiltrate the soil rather than running off
4. Sediment carried by runoff is also trapped behind the bunds, gradually rebuilding topsoil

Documented outcomes in Burkina Faso:

- Crop yields increased by **40–50%** in the first season after bund construction
- In Yatenga Province, stone bunds combined with zai pits have reclaimed an estimated **300,000 hectares** of degraded land since the 1980s
- Water table levels have risen by up to **3 metres** in some areas where stone bunds have been used widely — improving well water supply for domestic use
- Cost: approximately **\$50–100 per hectare** using local labour and stone — extremely cost-effective compared to other interventions

- Self-maintaining: communities maintain their own bunds as they directly benefit from improved harvests

FMNR — Farmer Managed Natural Regeneration

FMNR (Farmer Managed Natural Regeneration): A land restoration technique developed in Niger by Australian agronomist Tony Rinaudo (known as "The Man Who Stopped the Desert") in the 1980s. Rather than planting new trees (which are expensive and often fail to survive in harsh conditions), FMNR involves protecting and managing the natural regrowth of trees and shrubs from existing root systems that survived below-ground even when above-ground vegetation was destroyed by overgrazing, drought, or fuelwood cutting.

The FMNR method:

- Farmers select the strongest shoots emerging from dormant underground tree stumps and root systems (which can remain alive for decades)
- They prune competing weak shoots and protect the selected shoot from grazing animals
- Trees can grow to 2–3 metres within 3–4 years — far faster than planting new seedlings
- Farmers actively "manage" the trees — selectively harvesting branches for firewood and fodder while keeping the main trunk growing

Results in Niger (the "Re-greening of the Sahel"):

- An estimated **5 million hectares** of Niger's agricultural land has been restored using FMNR since the 1980s — one of the largest environmental transformations in Africa
- Approximately **200 million trees** have been regenerated on farmer fields across Niger
- Crop yields increased by **10–85%** depending on location — trees provide shade (reducing crop water stress), fix nitrogen (improving soil fertility), and reduce wind erosion
- Food security improved: surveys found that households in FMNR areas were more food secure, with extra food, fodder and income from trees
- Fuelwood supply improved dramatically — women (who collect fuelwood) reduced collection time from 2 hours to 30 minutes per day
- Cost: near-zero, as no seeds or planting required; farmers do the work themselves
- Technique has now been adopted across 24 countries in sub-Saharan Africa and beyond

Exam Tip — Management Strategy Comparison: FMNR vs Great Green Wall is an excellent comparison for evaluation questions. FMNR is: *bottom-up* (farmer-led), *low-cost* (near-zero), *fast* (results in 3–4 years), *locally appropriate* (uses species already adapted to local conditions), and *proven* (5 million hectares in Niger). The Great Green Wall is *top-down* (government/international), *expensive* (billions of dollars pledged), *slow* (only a small fraction complete after more than 15 years), and faces *political barriers* (coups, instability). This does NOT mean large-scale projects are useless — they can fund FMNR training and provide political will — but the lesson is that community-owned solutions tend to deliver better outcomes per dollar spent.

Exam Tip: For desertification questions, make sure you can distinguish between **physical** (climate) and **human** causes — many questions ask you to explain how BOTH contribute. The strongest answers also link causes together (e.g. population growth increases overgrazing, which combined with reduced rainfall creates a vicious cycle of degradation).

Key Comparison: In the exam you may be asked to compare management strategies. Low-tech, community-led approaches (zai pits, stone bunds, FMNR) are often *more effective in practice* than large-scale international projects (Great Green Wall) because they are owned by the community, use local knowledge, are cheap, and do not depend on political stability or international funding.

Drip Irrigation

Where crops are irrigated in semi-arid regions, the irrigation method itself can cause degradation. Flood and furrow irrigation deliver far more water than the plants use; the surplus either evaporates from the soil surface — drawing dissolved salts upward and leaving them behind as a white crust, a process called **salinisation** — or drains away wasted.

Drip irrigation delivers water through perforated pipes laid along the crop rows, releasing it slowly and directly at the base of each plant.

- **Efficiency:** Typically 80–90% of the water reaches the plant, against roughly 40–60% for flood irrigation, because almost nothing is lost to evaporation or runoff.
- **Prevents salinisation:** Because the soil is never saturated, salts are not drawn to the surface.
- **Fits a low-cost version:** NGOs promote gravity-fed "bucket kit" systems for smallholders, using a raised drum and plastic tubing with no pump and no electricity — a garden-sized system costs tens

rather than thousands of dollars.

- **Limitations:** Emitters block with silt and require filtration and regular cleaning; the initial cost of pipe is still beyond the poorest households; and it is suited to row crops and vegetables rather than to extensive cereal fields.

Eduqas names drip irrigation alongside magic stones and drought-tolerant crops as the NGO-scale techniques you should be able to describe and evaluate, so learn one concrete detail about each.

9. International Strategies

Local techniques restore land a field at a time. International strategies exist to coordinate, fund and scale them — and their record is more mixed.

Strategy	What it does	Evaluation
UNCCD (UN Convention to Combat Desertification)	The only legally binding international agreement on land degradation. Signatory countries produce national action programmes and report progress; it promotes the target of <i>land degradation neutrality</i> — no net loss of healthy land	Gives the issue a permanent institutional home and comparable data across countries, but has no enforcement power and depends on voluntary national action
The Great Green Wall	An African Union initiative to restore a belt of degraded land across the Sahel, now pursued as a mosaic of restoration projects rather than a single planted strip	See the detailed evaluation below
Development aid and climate finance	Donor funding for irrigation, seed distribution, training and restoration; climate funds increasingly treat restoration as adaptation	Can scale proven techniques quickly, but is vulnerable to shifting donor priorities and rarely lasts as long as ecological recovery takes
Emergency food aid	World Food Programme and NGO distributions during drought and famine	Saves lives, but treats the symptom. Poorly timed distributions can also undercut

		local farmers' prices, making the next harvest less viable
Research networks (e.g. CGIAR)	Breeding drought-tolerant sorghum, millet, cowpea and maize varieties and distributing seed through national systems	Directly raises yields under the conditions farmers actually face; depends on seed reaching smallholders, which is often the weak link

Evaluating the Great Green Wall (position as of 2025–26)

Target: 100 million hectares of degraded land restored, 250 million tonnes of carbon sequestered and 10 million green jobs created by 2030.

Progress: Roughly 18 million hectares restored and around 350,000 jobs created; more than US\$14 billion raised or pledged. The initiative's own assessment is that it stands at about 15% of its 2030 goal, and that meeting it would require restoring around 8.2 million hectares a year at roughly \$4.3 billion of annual funding.

What went wrong early: The original conception — a continuous 8,000 km wall of planted trees — produced very high seedling mortality, because trees were planted where rainfall could not sustain them and nobody had a reason to water them afterwards.

What changed: The programme now funds a mosaic of locally appropriate interventions — natural regeneration, water harvesting, improved grazing and agroforestry — rather than planting in straight lines. This is closer to what was already working in Niger and Burkina Faso.

What still blocks it: Incomplete monitoring data, coordination across many governments and NGOs, political instability and conflict in parts of the Sahel, and the long lag between money being pledged and work reaching the ground.

Judgement: The Great Green Wall is better understood as a funding and coordination mechanism for community-scale restoration than as a physical structure. Judged as a wall it has failed; judged as a way of channelling money and attention towards techniques that demonstrably work, it is making slow but real progress.